

Time: 90 minutes

Available marks: 62

Full marks: 60

Instructions

- 1. Fill in your details in the space provided below. You may also enter your student number on the top of each page in the space provided.
- 2. Answer all questions on this question paper in the boxes provided. You may use the backs of pages if you run out of space.
- 3. Write neatly and legibly, otherwise your answer will not be marked.
- 4. If you have read the instructions, draw a smiley face in the box at the bottom of this page. You will get one extra mark for doing so.
- 5. Show all work and reasoning unless otherwise instructed.
- 6. This test consists of 4 questions and 5 numbered pages. Make sure your question paper is complete.

Good luck! ✨⁺

Student Number

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Name & Last Name

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1. Let G be an abelian group. We define $D = \{(g, g) : g \in G\} \subseteq G \times G$. Use the First Isomorphism Theorem to prove that $(G \times G)/D \cong G$.

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2. (a) State the Second Isomorphism Theorem.
- (b) Let G be a group of order pq , where p and q are distinct primes. Suppose G has subgroups H and K such that $|H| = p$, $|K| = q$ and $K \triangleleft G$. Prove that $HK = G$.

HINT: Find $|HK|$.

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3. Let G be an abelian group (possibly infinite) with subgroups H and K such that $H \subseteq K$. Suppose $|G : K| = n$ and $|G : H| = m$ where $n \mid m$. Find $|K : H|$.

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4. Consider the group $G = \mathbb{Z}_{24}$ with subgroup $H = \langle [12] \rangle$.

(a) Determine all subgroups of $\mathbb{Z}_{24}$ and draw the lattice diagram.

(b) Use the Correspondence Theorem to draw the lattice diagram for $\mathbb{Z}_{24}/H$.

[10 + 5 = 15 marks]